

Numeric Computation with Numpy and SciPy

1 Numeric Computing in Python

- NumPy and SciPy in Python
- the computational ecosystem

2 Scientific Computing

- linear algebra
- differential equations

MCS 320 Lecture 40
Introduction to Symbolic Computation
Jan Verschelde, 3 August 2026

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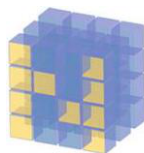
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NumPy and SciPy in Python



NumPy

defines vectors and matrices,
is a foundational project for the
Python scientific computing stack.



SciPy

provides fundamental algorithms
for scientific computing in Python.

`https://numfocus.org`

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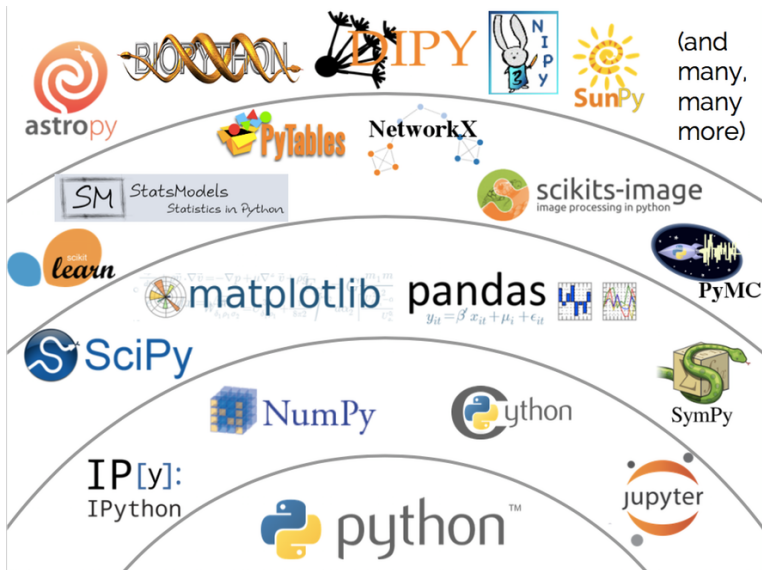
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The Computational Ecosystem

picture from the slides of Jake VanderPlas, 2015



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Linear Algebra

- NumPy provides vectorized functions, functions that take an entire vector on input and return a vector on output. Vectorized code speeds up Python code significantly.
- The backslash operator `\` as in $\mathbf{x} = A \backslash \mathbf{b}$ solves the linear system $A\mathbf{x} = \mathbf{b}$.
The solvers are implemented in optimized software libraries.
- Arrays are data structures, matrices are arrays with defined linear algebra methods.

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Differential Equations

a model of a pendulum

$$\frac{d^2}{dt^2}\theta(t) = -\frac{d}{dt}\theta(t) - g \sin(\theta(t)), \quad \theta(0) = \pi/10, \quad \theta'(0) = 0$$

Plotting both displacement θ and velocity θ' gives a phase portrait:

